

Cyber Security Internship Project Report

SOC-Based Log File Analyzer for Intrusion Detection

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1. Introduction

A Security Operations Center (SOC) is responsible for continuously monitoring, detecting, and responding to cybersecurity threats. Log analysis plays a crucial role in identifying malicious activities such as unauthorized access, brute-force attacks, and denial-of-service (DoS) attempts.

This project implements an advanced SOC-based log file analyzer that processes system and web server logs to detect and respond to potential security threats in real time.

2. Abstract

The objective of this project is to design and develop a log-based intrusion detection system that simulates SOC operations. The system parses Apache and SSH logs, detects suspicious patterns such as brute-force attacks, scanning

attempts, and DoS activities, and generates alerts.

It also integrates a threat intelligence mechanism to evaluate IP reputation and assigns risk scores to detected threats. The system supports real-time monitoring and generates structured reports for further analysis, making it useful for security monitoring and incident response.

3. Tools Used

- **Python** – Core programming language
- **Regex (Regular Expressions)** – Log parsing
- **Matplotlib** – Data visualization
- **JSON** – Report generation
- **OS Module** – File handling
- **Simulated Threat Intelligence** – IP reputation checking

4. Steps Involved in Building the Project

Step 1: Log Collection

Apache and SSH log files are used as input sources for analysis.

Step 2: Log Parsing

Regular expressions are used to extract relevant information such as IP address, request type, and status codes.

Step 3: Attack Detection

The system identifies:

- **Brute Force Attacks** – Multiple failed login attempts
- **DoS Attacks** – High number of requests from a single IP

- **Scanning Activity** – Multiple 404 (Not Found) responses

Step 4: Threat Intelligence Integration

The system checks whether an IP address is malicious using a predefined blacklist.

Step 5: Risk Scoring

Each detected threat is assigned a risk score based on:

- Type of attack
- IP reputation

Step 6: Alert Generation

Real-time alerts are displayed in the terminal when suspicious activity is detected.

Step 7: Report Generation

A structured report is generated in JSON format and stored in the reports/ folder for further analysis.

5. Features of the System

- Real-time log monitoring
 - Detection of multiple attack types
 - Threat intelligence integration
 - Risk-based alerting system
 - Automated report generation
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6. Conclusion

The SOC-based log file analyzer successfully demonstrates how real-world security systems monitor and detect cyber threats. It provides

hands-on experience in log analysis, intrusion detection, and incident response.

This project can be further enhanced by integrating real threat intelligence APIs, developing a graphical dashboard, and implementing automated response mechanisms, making it suitable for enterprise-level SOC environments.
